

Mechanical Explanatory Approach

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Starting Point of the Working Model

The mechanical explanatory approach is explicitly to be understood as a working model and thought experiment. It serves to vividly think through the hypothesis and to identify possible physical mechanisms. Therefore, it does not constitute a commitment that every sequence described herein must actually occur in nature in exactly this way.

The starting point of the working model is the fundamental assumption:

Universe = E

Matter and radiation are not regarded as ontologically distinct basic components, but rather as different states and organizational forms of the same fundamental energy and wave structure.

In this working model, the hypothesis assumes that during the formation of a black hole, the collapsing object segregates itself from the photons of its original universe. The collapsed object thus forms its own total physical system without the original external environment.

Through gravitational self-interaction and the associated contraction, energy can further concentrate or "charge up" in its interior. Possibly, further physical effects play a role in this process.

This energy is finally released outward in the form of photons or a photonic wave or field structure.

Possibilities regarding the Location of the White Hole

The hypothesis leaves various possibilities and details open in this regard:

- The white hole is located at the center of the black hole.
- The white hole is located alongside the original universe.
- The white hole forms a new location that is independent of the original universe.

In this context, the white hole does not necessarily have to be understood as a separate, static object. It could equally be a dynamic internal state or an ongoing physical process.

Possibilities regarding the Location of the New Universe

The location of the resulting new physical system also remains open:

- The new universe arises between the central white hole and the outer event horizon—that is, in the region that would be described as a black hole from the perspective of the original universe.

- The new universe overlaps with the original universe. Its emitted photons or wave structures possess different, possibly mirrored properties and evolve more or less independently of the original universe.
- The new universe arises at a completely new location, independent of the original universe. There, one of the aforementioned possibilities would again be conceivable without both universes overlapping spatially.

Possibilities regarding the Nature of the Emitted Energy

The white hole emits energy outward.

Within the working model, it is initially assumed that this energy consists of photons. An alternative possibility is that it represents a more fundamental form of energy whose smallest or most elementary excitation manifests to an observer as a photon.

Another possibility is that the emitted energy does not have to be understood as individual, mutually independent photons, but rather as a coherent electromagnetic or more general gauge field. In this case, the emitted field itself would be the fundamental structure. A photon could then appear as the smallest or most elementary excitation of this field.

Within this model, a photon could, for example, manifest as an individual excitation by having its state—such as with regard to polarization, spin, or another property yet to be specified—locally differ from the surrounding wave or field structure.

Which of these descriptions represents the actual underlying physical structure remains open.

Possibilities regarding the Spatial Distribution of Energy

The white hole could distribute the energy outward in different ways:

- The energy is initially released outward chaotically.
- The energy is released in ordered chains or radial structures.
- The photons are emitted in such a way that rather than creating individual, separated chains, a coherent three-dimensional photonic or field-like system is formed.

These possibilities are not necessarily independent of one another. An initially chaotic structure could, for example, transition into more ordered structures through interactions.

Possibilities regarding the Further Evolution of Energy

The further evolution of the emitted energy also remains open:

- Chaotically distributed energy states remain enclosed within the event horizon. The increasing or altered energy density appears from our perspective as expansion.
- The energy propagates outward, whereby the observed expansion corresponds to an actual spatial development.
- The radial chains propagate rectilinearly and unimpeded at c .
- The chains return to the center and form a closed wave. This can either maintain a constant size or grow over time.

- The photonic or field-like system remains approximately constant in its spatial extent, but changes within its structure.
- The photonic or field-like system grows at c .

Which of these possibilities applies would be one of the fundamental questions for a later mathematical description.

Possibilities regarding Energy Quantity and Energy Budget

The exact energy balance of the white hole remains open within the working model.

Possible scenarios are:

- The white hole generates, through gravitational self-charging, approximately or exactly the amount of energy that it emits outward. As a result, its total energy could remain approximately constant over the long term.
- The white hole emits energy without generating new energy to the same extent. Its available energy could thereby decrease over the long term.
- Due to the underlying dynamics, more energy could be supplied or become available to the white hole than it currently radiates, so that its total energy increases.
- The energy emitted by the white hole corresponds at the same time to the energy used for gravitational self-charging. This could create a closed loop, for example in the form of a closed wave or a photonic field.

Thus, it remains explicitly open whether the white hole actually continuously generates new energy or whether it is rather a closed or self-sustaining energy process. Likewise, it remains open whether the total energy of the system remains constant, decreases, or increases over the long term.

The Photonic Wave Structure

Within this hypothesis, the wave structure emanating from the white hole is itself the energy propagating outward.

The fundamental waves initially run radially and rectilinearly outward. Each individual wave is understood as a coherent structure made of photons.

The photons are thus not an additional object moving along a pre-existing wave. Rather, the photons, through their collective arrangement and dynamics, form the wave structure itself.

In this view, no conflict arises between "photon" and "wave." Instead, the photon denotes an elementary constituent or an elementary excitation within the photonic wave structure.

The fundamental wave structure initially propagates at c .

Due to the high density of the radial waves and their interactions, a high shared energy density can arise locally. This energy and momentum distribution contributes to the energy-momentum tensor $T_{\mu\nu}$ and generates spacetime curvature in accordance with General Relativity.

This gravitational feedback can in turn act back on the wave structure itself.

Local State Changes and Pattern Transmission

The individual photons of a wave are considered coupled to one another within the working model.

A local interaction can thereby alter the state of the wave structure in a specific spatial region. Due to the assumed coupling, each subsequent photon passing through this region can adopt the same altered state as the photon previously located there.

In this way, for example, a local oscillation, polarization change, or other state change could propagate along the wave.

Crucially, a stable pattern does not have to remain permanently bound to the same photons or to a specific wave line. A pattern can transfer to neighboring waves within a dense three-dimensional wave field or shift its state onto other waves.

The individual photons continue to move according to the fundamental wave evolution, while the stable pattern propagates through the wave structure as an organized state.

Whether the individual waves are actually coupled to one another and thereby form a coherent three-dimensional field remains an open question.

Polarization, Spin, and Further Properties

Polarization could play an important role in the coherence and stability of such patterns within this model.

Spin or other quantized properties of a photon could also possibly be linked to the geometry and dynamics of the emerging wave states.

It also remains open whether certain stable structures can arise only through a specific combination of polarization, spin, phase, frequency, and spatial arrangement.

These details are part of a later mathematical examination and are not initially fixed by the working model.

Gravitational Feedback and Self-Binding

With a very large number of closely neighboring waves and a corresponding number of mutually interacting states, the shared energy density would rise sharply.

This could trigger the previously described gravitational self-interaction.

Schematically:

- **Wave structure → interaction → high shared energy density → $T_{\mu\nu}$ → spacetime curvature → gravitational feedback → altered wave structure**

If a portion of the energy within this wave structure becomes gravitationally bound, this fraction of energy is no longer available in the same way for free radial evolution.

This could give rise to a stable or at least long-lived organized energy state with rest energy or rest mass.

It is not assumed that individual photons simply lose their speed of light or are "slowed down" by inertia. Rather, the organization of energy changes within the entire coupled wave and spacetime structure.

The larger and more stable such a bound state becomes, the more strongly its energy-momentum distribution in turn influences the surrounding spacetime and thus neighboring waves.

Schematically:

- **Wave structure** → **interaction** → **energy concentration** → **$T_{\mu\nu}$** → **spacetime curvature** → **self-organization / gravitational binding** → **stable energy state**

Whether such a self-bound electromagnetic or photonic field configuration can actually be stable or long-lived would be a question to be investigated mathematically, for example within a coupled Einstein-Maxwell framework.

Stable Oscillation Patterns and Atomic Nuclei

Within this hypothesis, stable physical structures can be understood as collective oscillation and organization patterns of multiple coupled photonic states.

In this working model, an atomic nucleus would be, for example, one of the smallest stable complex organizational forms: a collective oscillation pattern of at least two or a few mutually coupled fundamental states.

In this context, coupling is not to be understood as an additional connection between two complete matter particles. Rather, it would be a shared state of the participating photonic wave structure, for example through shared polarization, oscillation, and other properties.

The structure is not permanently bound to specific photons or to a specific wave line. The pattern can transfer to neighboring waves within the dense wave field and thereby move spatially.

Thus, the stable structure itself would be the physically relevant object—not the identity of the individual photons of which it consists at a given moment.

The surrounding waves do not each have to develop an independent active oscillation themselves. Rather, they can be passively influenced, deformed, or deflected by the energy-momentum distribution and gravitational interaction with the stable structure.

It remains open whether such a surrounding structure actually represents an independent stable form of oscillation or is predominantly a passive gravitational or geometric reaction of neighboring waves.

Chemical Structures as a Higher Level of Description

If stable oscillation and organization patterns form the basis of material structures, known representations in chemistry could be understood at a higher level of description as simplified representations of these deeper structures.

Thus, for example, chemical structural formulas might not represent the fundamental physical structure itself, but rather abstract the relationships between stable energy and oscillation structures.

Nodes and bonds in chemical representations could possibly represent a simplified to approximately precise mapping of the underlying physical connection structure.

Different chemical compounds in this interpretation would be different stable organizational patterns of the same fundamental energy and wave structure.

This consideration is explicitly speculative and does not represent an established physical statement.

Further Illustrative Comparisons

The basic idea can also be illustrated using simpler processes.

Spilled water spreads out and forms several separate puddles. A glass shatters and forms many new structures out of an originally connected structure.

The emergence of a new organism can also be understood within this purely structural perspective as the emergence or splitting off of a new, complex organizational pattern from an original pattern.

These examples are not physical proofs of the hypothesis. They are merely intended to illustrate how new stable structures can arise from a connected system through interaction, division, and reorganization.

In all cases, within the hypothesis, these would be changes and splittings within an overarching energy and/or wave pattern.

Approach to Special Relativity

Within this hypothesis, the outward propagation of fundamental energy waves at c is viewed as the fundamental temporal and spatial evolution of the universe.

This wave evolution thus forms, in a sense, the physical foundation of universal time, spatial evolution, and all energy.

Motion is understood within this working model as a redistribution of this fundamental energy evolution.

A moving system would accordingly bind a portion of its available evolutionary dynamics into its spatial motion. Correspondingly, less evolutionary capacity would be available for the system's internal proper temporal evolution.

The closer a system approaches the speed of light (c), the more its proper time would be reduced compared to universal temporal evolution.

In the limiting case $v \rightarrow c$, internal proper temporal evolution in this qualitative picture would approach zero.

The time dilation of Special Relativity is thus interpreted within the hypothesis as a consequence of the distribution of energy evolution between spatial motion and proper temporal evolution.

This representation is initially exclusively qualitative. Whether the known relativistic relationships—in particular the Lorentz transformations and the relativistic energy-momentum relation—can actually be derived mathematically from this remains an open question.

The Big Bang as an Ongoing Process of Emergence

Within this hypothesis, the Big Bang might not be understood as a single, fully concluded event, but rather as an ongoing process of emergence and evolution.

Around the central white-hole state, radial photon waves could initially oscillate extremely strongly and complexly, thereby forming a highly excited wave structure.

This region would represent the state we perceive as the early Big Bang state when looking into the past.

This zone of emergence and evolution would not be closed off, but could be continuously pushed further outward by the ongoing energy flow of the white hole.

New energy and new wave states would be continuously introduced into the system, while already formed structures continue to evolve and move further outward or alter their states.

How long the central white-hole state had already been active before our observable region left this original zone of emergence would be unknown within this hypothesis.

The evolution could have originated from an initially spherically symmetric situation. However, it does not follow that all subsequent regions must possess the same state of development.

Individual waves and wave regions can be influenced to varying degrees by different interactions, gravitational influences, and the formation of stable structures.

As a result, different developmental states could emerge, even though all originated from the same initially spherically symmetric evolution.

The Big Bang in this sense would not be a process that lies entirely in the past, but an ongoing process in which the white-hole state introduces new energy and new wave states into the system, causing the zone of emergence to further evolve or shift outward.

Recursive Development

From the entire working model, the possibility of a recursive process finally emerges:

- **Energy → structure → concentration → black hole → possible white-hole state → energy propagation → new structures → renewed concentration → possibly new black holes**

Thus, it is conceivable that a black hole not only represents the end of a previous evolutionary process, but could, under certain conditions, form the starting point of a new physical total system.

Whether a new universe actually arises from this, whether every black hole undergoes such a process, and whether such a process in turn yields new black holes remains open.

Likewise, it remains open whether the energy resulting from this is actually newly generated or whether it is ultimately a closed cycle of pre-existing energy.

Final Contextualization

The mechanical explanatory approach does not attempt to definitively fix every single process.

Rather, it serves to vividly illustrate a possible path from the fundamental energy of the white hole via photonic wave structures, interactions, stable oscillation and organization patterns, and gravitational feedback loops all the way to the observable structures of the universe.

The various possibilities mentioned—such as regarding the position of the white hole, spatial evolution, the nature of the gauge field, the significance of the photon, the energy balance, or the concrete emergence of stable structures—are therefore components of the thought experiment and not definitive commitments of the hypothesis.

Individual ideas among these may prove to be incompatible with a consistent physical description upon later mathematical investigation and would then have to be discarded or replaced.

The core of the working model remains independent of this:

A collapsed object can form a black hole and possibly simultaneously yield a white-hole state or a corresponding internal process of emergence. This process leads to a photonic or field-like propagation of energy.

Through interactions, oscillations, polarization, gravitational feedback, and other potentially relevant properties, stable organizational patterns can arise from this. Within the hypothesis, these patterns form the basis of observable physical structures.

Which of these mechanical concepts are actually physically possible can ultimately only be decided by a mathematically consistent formulation and comparison with known observations.

AI Assistance Statement

Generative AI tools, including ChatGPT (OpenAI), were used as research and writing assistants during the development of this hypothesis. Their use included discussion of physical concepts, critical examination of assumptions, organization and revision of the manuscript, and language editing.

The author remains solely responsible for the scientific content, assumptions, interpretations and conclusions presented in this work. AI-generated suggestions were critically evaluated and do not constitute independent scientific validation of the hypothesis.